

Less Traffic, Higher Tolls: Optimal Congestion Pricing under Remote Work and Traveler Heterogeneity

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Abstract

This paper studies how a persistent shift toward remote work changes optimal freeway congestion prices. I combine hourly station traffic from the California Performance Measurement System with tract-level commuting patterns, work-from-home shares, industry composition, population, and wages for five large California counties. Station traffic is allocated to residential tracts using fixed pre-pandemic commuting-route exposure, producing an estimated tract-level freeway VMT measure that aggregates exactly to observed PeMS VMT. A shift-share design based on predetermined local industry shares and national leave-California-out changes in industry remote work implies that a 10 percentage point increase in working from home reduces tract freeway VMT by 1.43 percent. A capacity-normalized BPR technology estimated by Poisson pseudo-maximum likelihood yields a congestion curvature of 3.27. These inputs produce three main pricing findings. First, holding the value of time fixed, the demand shift lowers average optimal tolls by 2.4–4.7 percent. Second, allowing the value of time to move with local wages reverses the result: optimal tolls rise by 8.0–11.4 percent. Third, income heterogeneity changes the meaning of distributional comparisons. Lower-income tracts supply 77–84 percent of modeled traffic and bear most of the marginal external damages in the common Pigouvian calculation. Under a constrained policy that imposes the same time-equivalent charge on both groups, however, the monetary rate paid by higher-income travelers is 1.6–2.0 times larger because their value of time is higher. Distributional conclusions therefore depend on whether the object of interest is a group's role in and exposure to external delay or the monetary rate it pays.

Keywords: Congestion pricing, remote work, vehicle miles traveled, value of time, shift-share instruments, distributional incidence

JEL Classification: R40, R41, D62, H23

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1 Introduction

Remote work can reduce commuting traffic, but less congestion does not necessarily imply a lower optimal congestion tax. The tax compensates for the delay that an additional driver imposes on other road users. Its monetary value depends on both the amount of delay and how travelers value their time. A decline in physical delay can therefore coincide with a higher optimal tax if the dollar value of time rises sufficiently. Differences in remote-work opportunities and wages across travelers also raise the question of how this adjustment should be reflected in the rates paid by different income groups.

My paper asks how the shift toward remote work changes optimal freeway congestion taxes when travel demand and the value of time differ across income groups. I study Los Angeles, Orange, San Diego, San Francisco, and Santa Clara counties, comparing 2018–2019 with 2021–2022. The analysis builds on the principle that the optimal congestion charge equals the cost of the delay imposed on other travelers ([Pigou, 1920](#); [Small, Verhoef, & Lindsey, 2024](#); [Vickrey, 1969](#)). I connect this pricing condition to estimated remote-work demand shifts and observed wage changes, and examine both a common monetary toll and a policy that gives income groups the same charge expressed in units of travel time.

My first contribution is to show empirically how the monetary valuation of delay can reverse the pricing implications of a reduction in traffic demand. I separate changes in marginal physical delay from changes in its dollar value, allowing the value of time to vary with local wages. This connects the observed traffic adjustment to the economic cost that determines the tax. Related empirical work estimates congestion charges from traffic conditions and behavioral responses ([Kreindler, 2024](#); [Yang, Purevjav, & Li, 2020](#)). My analysis brings changes in time valuation into that calculation, drawing on evidence that values of time differ across people and settings ([Bento, Roth, & Waxman, 2024](#); [Lam & Small, 2001](#); [Small, 2012](#)).

My second contribution is an empirical pricing framework that accounts for differences in remote-work exposure, demand responses, traffic use, and the value of time across income groups. I estimate demand shifts separately for lower- and higher-income tracts and solve their demands jointly because both groups create congestion on the same freeway network. This allows a demand change in one group to affect the delay experienced by everyone. The analysis extends the heterogeneous-user pricing framework of [Verhoef and Small \(2004\)](#) by incorporating estimated income-specific remote-work responses. It also distinguishes the external delay borne by each group from the monetary rate that group pays, a distinction relevant to the distributional analysis of road pricing ([Eliasson & Mattsson, 2006](#); [van den Berg & Verhoef, 2011](#)).

My third contribution is to compute a common time-equivalent congestion charge using empirical estimates of traffic conditions and income-specific demand shifts. Under this restriction, each group's monetary toll divided by its value of time is the same, so a higher value of time implies a higher dollar-per-mile rate. [Pandey, Alamri, Khoury, and Bakre \(2022, Proposition 1\)](#) provide a theoretical precedent for this proportionality rule in an express-lane model. To my knowledge, my paper is the first to combine empirical estimates of congestion technology and remote-work demand shifts to compute the optimal common time-equivalent charge. I calculate both this charge and the corresponding monetary rates for each income group, and examine how they change with remote work and wages.

I combine hourly freeway detector observations with commuting routes and residential tract characteristics. The California Performance Measurement System records freeway traffic, while the Longitudinal Employer-Household Dynamics Origin-Destination Employment Statistics (LODES) data identify where workers live and work. I use fixed pre-pandemic commuting-route exposure to allocate station traffic across residential tracts. The allocation preserves observed county totals. Census data provide remote-work shares, industry composition, population, and earnings.

I estimate the effect of remote work on traffic using differences in tracts' exposure to national changes in industry-specific remote work. Local work-from-home shares may respond to labor-market conditions, residential selection, and transportation shocks. I therefore instrument them with 2019 tract industry shares interacted with national changes in industry remote work calculated outside California, following the shift-share literature ([Borusyak, Hull, & Jaravel, 2022](#); [Goldsmith-Pinkham, Sorkin, & Swift, 2020](#)). Tract and county-by-year fixed effects compare differently exposed tracts within the same county and year. I combine the estimated demand shifts with a county-specific relationship between traffic and travel time and wage-based values of time to solve for optimal hourly taxes. The price elasticity of demand is calibrated, while the response to remote work is estimated.

For the time-of-day analysis, I estimate separate congestion curves for morning peak, afternoon peak, and off-peak hours to account for differences in traffic conditions over the day. These curves determine the corresponding hourly tax profiles. Taxes are largest around the commuting peaks and effectively zero during some off-peak hours. With contemporaneous wages, the post-period hourly tax schedule generally lies above its pre-period counterpart.

The main pricing result is that optimal monetary taxes rise despite the reduction in congestion when the value of time follows contemporaneous wages. Holding the value of time at its 2018 level, the demand shift lowers average optimal taxes by 2.4–4.7 percent across the five counties. Allowing wages to change reverses the comparison, with taxes rising by 8.0–11.4 percent. The decomposition attributes a reduction of 7.3–9.4 percent of the initial tax to lower marginal physical delay and an increase of 17.0–18.8 percent to the higher value of time. The valuation component is larger, so the net monetary tax rises.

The heterogeneous-agent results show that the groups bearing most external-delay damage need not face the highest monetary rates under a common time-equivalent charge. Lower-income tracts supply 77–84 percent of modeled traffic and bear most of the marginal external-delay cost in the common monetary toll calculation. Under the constrained time-equivalent

policy, higher-income travelers pay dollar-per-mile rates 1.6–2.0 times larger because their value of time is higher. The same group can therefore account for most traffic and bear most of the added delay while paying the lower monetary rate. These comparisons describe traffic use, exposure to external delay, and toll rates; they do not measure net welfare effects by income group.

The demand estimates imply that a 10 percentage point increase in working from home reduces allocated tract freeway vehicle miles traveled by 1.43 percent. Under the design’s identifying assumptions, this is a causal effect of remote work on traffic demand. The estimate connects changes in work arrangements to the demand shifts used in the pricing exercise. Existing research examines how remote work changes commuting and the locations of homes and workplaces (Delventhal, Kwon, & Parkhomenko, 2022; Loo & Huang, 2022), while differences in its feasibility across industries generate unequal exposure to those changes (Dingel & Neiman, 2020). My analysis carries the estimated traffic response into congestion-tax calculations for travelers sharing the same road network.

Section 2 develops the pricing model. Section 3 describes the data and traffic allocation. Section 4 presents the congestion and demand specifications. Section 5 reports the pricing results and income-group comparisons. Section 6 discusses the implications for congestion-tax design. The appendices provide the theoretical derivations and supplementary evidence.

2 Model

Computing an optimal congestion charge requires three objects: a technology that maps traffic into travel time, a demand curve that maps generalized cost into freeway use, and a value of time that converts delay into dollars. County is indexed by c , hour of the weekly profile by h , calendar year by t , and flow by F in passenger-car-equivalent vehicle miles per hour.

2.1 Congestion technology and travel costs

Throughout this section, F denotes a generic traffic-flow level. When evaluating the model for county c , hour h , and year t , I use F_{cht} . The baseline congestion function $T_c(\cdot)$ is county-specific; travel time varies across hours and years through changes in traffic flow.

Let S_c^0 denote free-flow speed, measured in miles per hour, and K_c the engineering capacity of the observed freeway network. Free-flow speed is the speed vehicles can maintain when traffic is light and other road users do not slow them down. Following the functional form introduced by [U.S. Bureau of Public Roads \(1964\)](#), the BPR travel-time function per mile is

$$T_c(F) = \frac{1}{S_c^0} \left[1 + \alpha_c \left(\frac{F}{K_c} \right)^\beta \right], \quad (1)$$

where $\alpha_c > 0$ is a county-specific scale parameter and $\beta > 0$ controls curvature. The ratio F/K_c is dimensionless, and $T_c(F)$ is measured in hours per mile. Differentiation gives

$$T'_c(F) = \frac{\alpha_c \beta}{S_c^0 K_c} \left(\frac{F}{K_c} \right)^{\beta-1}. \quad (2)$$

Following the standard distinction between private and social congestion costs ([Small et al., 2024](#); [Walters, 1961](#)), let $TSC_{cht}(F, T)$ denote total social travel-time cost, measured in dollars per hour, as a function of traffic flow F and travel time per mile T . This function adds the time costs borne by all travelers in a county-hour-year. Because network travel time depends on flow, total cost along the congestion curve is $TSC_{cht}(F, T_c(F))$. Before imposing a functional form for costs, the chain rule gives

$$\frac{dTSC_{cht}(F, T_c(F))}{dF} = \underbrace{\frac{\partial TSC_{cht}}{\partial F}}_{\text{direct cost}} + \underbrace{\frac{\partial TSC_{cht}}{\partial T} \times T'_c(F)}_{\text{congestion effect}}. \quad (3)$$

The partial derivatives are evaluated at $(F, T_c(F))$, holding the other cost determinants fixed.

The first term measures the direct cost of additional flow with travel time per mile unchanged. The second captures the change in total cost when additional flow raises travel time for existing traffic. Their sum is marginal social cost, measured in dollars per mile.

In the homogeneous-agent baseline, all travelers have the same value of time VOT_{ct} , measured in dollars per hour. The cost function takes the specific form

$$TSC_{cht}(F, T) = VOT_{ct} \times F \times T. \quad (4)$$

Evaluating this function at the flow and corresponding travel time for county c , hour h , and year t gives

$$TSC_{cht}(F_{cht}, T_c(F_{cht})) = VOT_{ct} \times F_{cht} \times T_c(F_{cht}).$$

Holding VOT_{ct} fixed, the first term in Equation (3) is therefore the private marginal cost (PMC) of one vehicle mile, or the time cost borne by its driver,

$$PMC_{cht}(F) = VOT_{ct} \times T_c(F) = \frac{VOT_{ct}}{S_c^0} \left[1 + \alpha_c \left(\frac{F}{K_c} \right)^\beta \right]. \quad (5)$$

The second term is the marginal external congestion cost ($MECC$), the value of the extra delay imposed on all existing traffic. Marginal social cost (MSC) includes both components,

$$MSC_{cht}(F) = PMC_{cht}(F) + MECC_{cht}(F), \quad (6)$$

$$MECC_{cht}(F) = VOT_{ct} F T'_c(F) = \frac{VOT_{ct}}{S_c^0} \alpha_c \beta \left(\frac{F}{K_c} \right)^\beta. \quad (7)$$

Thus $MECC$ is measured in dollars per mile. Appendix A derives Equations (6) and (7) from the driver's problem and the planner's problem.

I set the value of time to one-half of the local hourly wage,

$$VOT_{ct} = \theta w_{ct}, \quad \theta = 0.5, \quad (8)$$

following the conventional benchmark summarized by [Small \(2012\)](#). The factor $\theta = 0.5$ is held fixed across all policy exercises. Wages are either held at their county-specific 2018 levels or updated by county and year. In the heterogeneous-agent extension, wages and values of time also differ between the lower- and higher-income groups within each county.

2.2 Demand, equilibrium, and the Pigouvian tax

Let p_{cht} denote the driver's private cost of travel time and tolls for one additional mile of freeway travel in county c , weekly hour h , and year t , measured in dollars per mile. Travel time is converted into dollars using the value of time, and the monetary toll is added to this amount,

$$p_{cht} = PMC_{cht}(F_{cht}) + \tau_{cht} = VOT_{ct} \times T_c(F_{cht}) + \tau_{cht}.$$

Here VOT_{ct} is the value of time in dollars per hour, $T_c(F_{cht})$ is travel time in hours per mile at traffic flow F_{cht} , and τ_{cht} is the toll in dollars per mile. Thus p_{cht} is neither the toll alone nor the marginal social cost. Without a toll, it equals the driver's travel-time cost. Fuel and other vehicle operating costs are excluded from p_{cht} . In the demand regression, county-by-year fixed effects absorb their effects on traffic to the extent that those effects are common to tracts in the same county and year (Equation (28)).

For the policy simulations, I adopt the constant-elasticity travel-demand specification used by [Yang et al. \(2020\)](#),

$$F_{cht} = A_{cht} p_{cht}^{-\eta}, \quad \eta = 0.1, \quad (9)$$

where A_{cht} is a positive demand shifter and $-\eta$ is the elasticity of freeway use with respect to generalized monetary cost. This parsimonious form separates changes in demand from movements along the demand curve. Remote work and population shift A_{cht} , while η governs the response to a change in price. It implies $\partial \ln F_{cht} / \partial \ln p_{cht} = -\eta$, holding A_{cht} fixed, and keeps demand positive at positive prices. Constant elasticity is a maintained functional-form assumption, not a result of the remote-work estimates. I set $\eta = 0.1$ as a common calibration across counties, hours, and income groups; the empirical design estimates demand shifts rather than this price elasticity.

With a monetary toll τ_{cht} , private equilibrium satisfies

$$F_{cht} = A_{cht} [PMC_{cht}(F_{cht}) + \tau_{cht}]^{-\eta}. \quad (10)$$

Following the standard welfare-maximizing congestion-pricing condition ([Small et al., 2024](#), Section 4.1), the planner equates inverse demand with marginal social cost. Efficient flow F_{cht}^* therefore solves

$$F_{cht}^* = A_{cht} [MSC_{cht}(F_{cht}^*)]^{-\eta}, \quad (11)$$

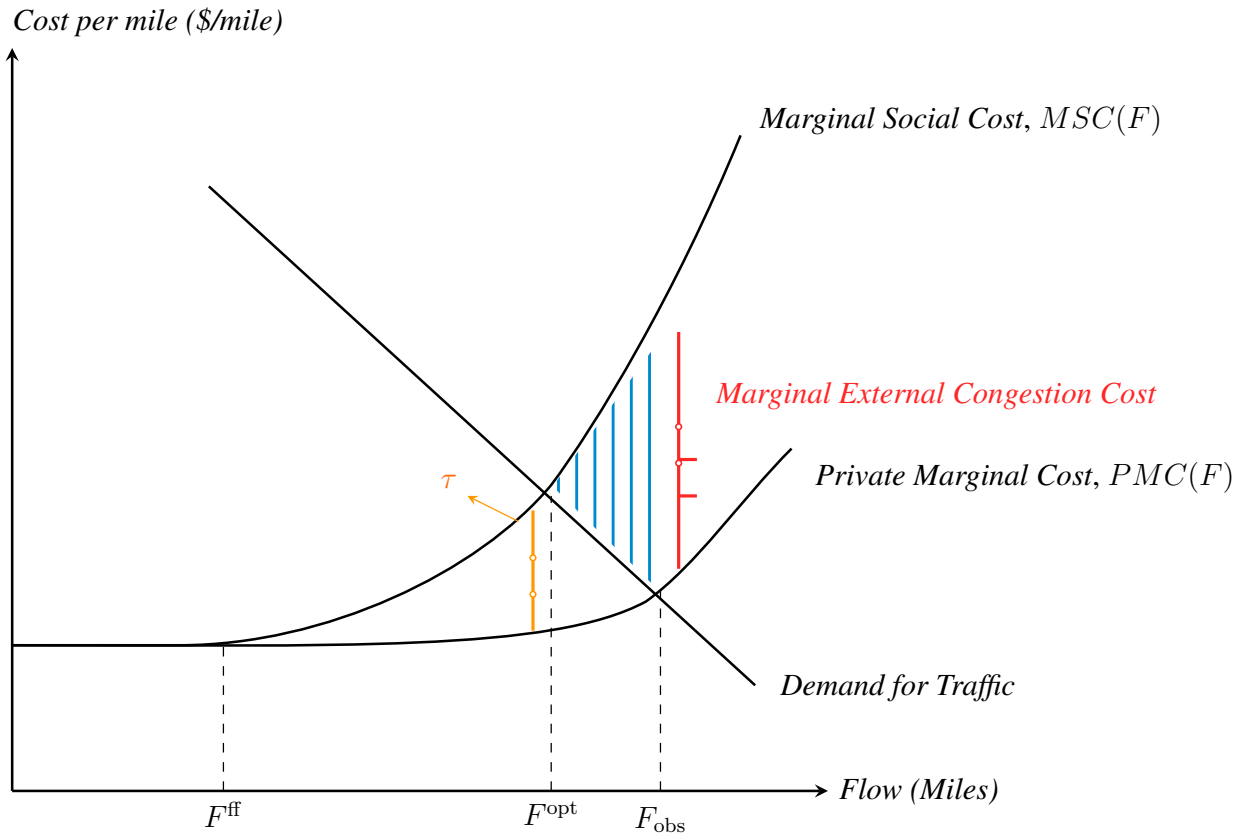
and is decentralized by the Pigouvian tax

$$\tau_{cht}^* = MECC_{cht}(F_{cht}^*). \quad (12)$$

The tax is not chosen to make traffic as small as possible. It is the value of the delay imposed on other travelers at the efficient flow. Raising the tax beyond Equation (12) would remove

trips whose willingness to pay exceeds their social cost and would lower welfare. Figure 1 illustrates this result. Without a toll, demand intersects private marginal cost at F_{obs} . Marginal social cost lies above private marginal cost by the marginal external congestion cost, so the efficient intersection occurs at the lower flow F^{opt} . The vertical gap between the two cost curves at F^{opt} is the optimal toll, and the hatched wedge is the deadweight loss from unpriced congestion. Appendix B provides the formal first-order condition.

Figure 1. Private and Marginal Social Costs of Congestion

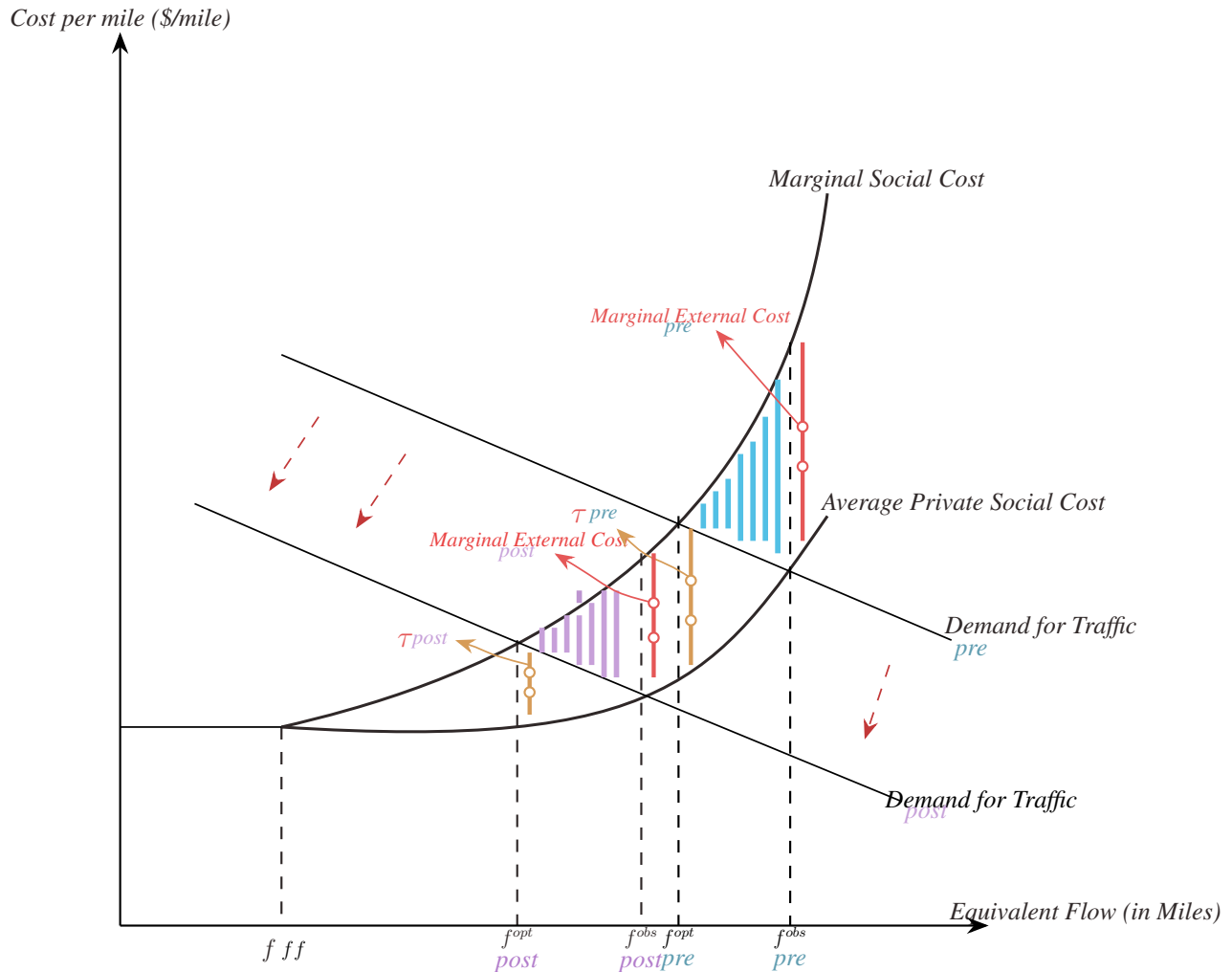


Notes: The unpriced flow F_{obs} occurs where demand intersects private marginal cost. Efficient flow F^{opt} occurs where demand intersects marginal social cost. The optimal per-mile toll equals marginal external congestion cost at the efficient flow. The hatched region represents the deadweight loss from unpriced congestion.

Figure 2 applies the same logic before and after a decline in travel demand. Holding the congestion technology and the value of time fixed, the inward demand shift moves both

the unpriced and efficient intersections to lower flows. Because marginal external delay is lower at the new efficient flow, the optimal toll falls. This is the mechanism isolated by the fixed-value-of-time counterfactual; the empirical analysis subsequently allows wages, and therefore the monetary value attached to delay, to change as well.

Figure 2. Optimal Congestion Pricing Before and After a Demand Shift



Notes: A downward demand shift reduces unpriced and efficient flow. Holding the congestion technology and value of time fixed, the marginal external cost and optimal tax fall. The empirical exercise separately allows the value of time to change.

2.3 Toll incidence and congestion-cost offset

Suppressing subscripts, write the perceived price as $p(\tau) = PMC(F(\tau)) + \tau$. Taking logs of the private-equilibrium condition gives

$$\ln F = \ln A - \eta \ln[PMC(F) + \tau]. \quad (13)$$

Differentiating both sides with respect to the toll,

$$\frac{1}{F} \frac{dF}{d\tau} = -\frac{\eta}{PMC(F) + \tau} \left[PMC'(F) \frac{dF}{d\tau} + 1 \right]. \quad (14)$$

Collecting the terms that contain $dF/d\tau$ gives

$$\frac{dF}{d\tau} = -\frac{\eta F}{PMC(F) + \tau + \eta F PMC'(F)} < 0. \quad (15)$$

The toll reduces flow, which improves traffic conditions and lowers private time cost:

$$\frac{dPMC}{d\tau} = -\frac{\eta F PMC'(F)}{PMC(F) + \tau + \eta F PMC'(F)} < 0. \quad (16)$$

Accordingly,

$$0 < \frac{dp}{d\tau} = \frac{PMC'(F) + \tau}{PMC(F) + \tau + \eta F PMC'(F)} \leq 1. \quad (17)$$

The perceived price rises less than dollar-for-dollar because part of the charge is returned to drivers through lower congestion. This does not imply that an arbitrarily large tax is efficient. Equation (17) is an incidence result at any feasible tax; Equation (12) is the welfare condition selecting the tax. At the optimum,

$$\tau^* = F^* PMC'(F^*) = MECC(F^*), \quad (18)$$

so the pass-through expression becomes

$$\begin{aligned}\left.\frac{dp}{d\tau}\right|_{\tau=\tau^*} &= \frac{PMC(F^*) + MECC(F^*)}{PMC(F^*) + MECC(F^*) + \eta MECC(F^*)} \\ &= \frac{MSC(F^*)}{MSC(F^*) + \eta MECC(F^*)} \in (0, 1).\end{aligned}\tag{19}$$

Thus the optimal toll is determined by the marginal external cost at the efficient flow, while Equation (19) describes how a marginal change in that toll is divided between a higher perceived price and lower congestion cost. This distinction matters for interpretation: a one-dollar increase in the posted toll raises the driver's generalized price by less than one dollar because the induced reduction in traffic returns part of the charge as time savings. Pass-through therefore describes the incidence of a marginal tax change, not whether the tax itself is welfare maximizing.

2.4 Decomposition of the optimal congestion tax

The optimal tax equals marginal external congestion cost at the efficient flow. To distinguish physical delay from its monetary valuation, define

$$g_c(F) \equiv FT'_c(F), \quad \tau_{cht}^* = VOT_{ct} g_c(F_{cht}^*).\tag{20}$$

The function $g_c(F)$ is marginal external delay in hours per mile. It measures the additional delay imposed on existing travelers by one more unit of traffic, as derived in Appendix A. Differentiating marginal external congestion cost with respect to flow and the value of time gives the continuous decomposition

$$d\tau_{cht}^* = \underbrace{VOT_{ct} g'_c(F_{cht}^*) dF_{cht}^*}_{\text{congestion component}} + \underbrace{g_c(F_{cht}^*) dVOT_{ct}}_{\text{value-of-time component}}.\tag{21}$$

Here $g'_c(F) = T'_c(F) + FT''_c(F)$. The first term captures how a small change in efficient traffic changes the tax when the value of time is held fixed. The second captures how a small change in the value of time changes the dollar valuation of external delay when traffic is held fixed.

For the empirical before–after comparison, bars denote the reported county-period tax and value of time. The corresponding effective delay index is $g_{cr} \equiv \bar{\tau}_{cr}/\overline{VOT}_{cr}$, where $r \in \{pre, post\}$. Suppressing the county subscript, the exact discrete decomposition is

$$\Delta\bar{\tau} = \underbrace{\overline{VOT}_{pre}(g_{post} - g_{pre})}_{\text{congestion component}} + \underbrace{g_{post}(\overline{VOT}_{post} - \overline{VOT}_{pre})}_{\text{value-of-time component}}. \quad (22)$$

The first term changes the effective delay index from its pre-period to its post-period value while holding the value of time at its pre-period level. The second values the post-period index using the change in the value of time. Evaluating the index at its post-period value in the second term includes the interaction between the two changes, so the decomposition is exact for finite before–after differences.

3 Data

3.1 PeMS traffic and the observed freeway network

PeMS records hourly flow, VMT, speed, lane counts, and station location for California freeways. I use mainline and HOV stations in the five study counties. Flow in passenger-car equivalents (PCE), expressed as vehicle miles traveled (VMT) per hour, is

$$F_{cht} = VMT_{cht}^{passenger} + 1.5 VMT_{cht}^{truck}. \quad (23)$$

The factor 1.5 converts truck VMT into passenger-car equivalents. It is a transparent engineering benchmark rather than an estimated parameter.

Free-flow speed is estimated at the station level using low-flow observations in 2018 and 2019, requiring speeds between 50 and 90 miles per hour and adequate detector coverage. Mainline capacity per lane follows the standard engineering schedule $1,700 + 10S_s^0$ passenger cars per hour per lane when $S_s^0 \leq 70$ mph, capped at 2,400 above 70 mph. HOV capacity is set to 1,650 passenger cars per hour per lane.¹ Station capacity is multiplied by the number of lanes and the PeMS coverage length; county capacity K_c is the sum over stations, measured in PCE-VMT per hour, the same units as F_{cht} . Table 1 reports the resulting network inputs.

Table 1. Estimated Congestion Technology and Engineering Inputs

County	$\hat{\alpha}_c$	S_c^0 (mph)	K_c (million PCE-VMT/h)	Stations	Lane-miles
Los Angeles	2.583	66.85	10.407	2,623	4,562.9
Orange	1.649	67.37	4.402	1,667	1,950.7
San Diego	2.350	67.19	5.533	891	2,388.5
San Francisco	3.260	66.82	0.406	62	171.6
Santa Clara	2.720	66.26	2.497	554	1,058.6

Notes: The preferred PPML specification estimates county-specific scale parameters and a common curvature parameter, $\hat{\beta} = 3.270$ (HC1 standard error 0.019), using 175,180 county-hour observations. S_c^0 is the capacity-mile-weighted harmonic mean of station free-flow speeds. K_c sums station engineering capacity times PeMS station coverage length over mainline and HOV detectors. Capacity is measured in passenger-car-equivalent VMT per hour (PCE-VMT/h); trucks receive a factor of 1.5.

These capacity measures describe the freeway mileage represented by the detector network, not the administrative capacity of every road in the county. This alignment is important: the numerator F_{cht} and denominator K_c cover the same observed network. Unlike these engineering inputs, the county scale parameters $\hat{\alpha}_c$ in Table 1 are estimated jointly with the common curvature β by Poisson pseudo-maximum likelihood using Equation (27), presented in Section 4.

¹The capacity benchmarks follow the Federal Highway Administration’s HPMS guidance and Freeway Management Handbook.

3.2 Allocating station VMT to residential tracts

PeMS does not identify the residential origin of a vehicle. I therefore allocate observed station VMT to origin census tracts using pre-pandemic LODES commuting pairs and modeled freeway routes:

$$\widehat{VMT}_{it} = \sum_{s \in \mathcal{S}_{c(i)}} \left[\frac{\sum_j \bar{N}_{ij}^{pre} A_{ijs}}{\sum_{k \in \mathcal{I}_{c(i)}} \sum_j \bar{N}_{kj}^{pre} A_{kjs}} \right] \left[\sum_{h \in t} Flow_{sh} L_s \right]. \quad (24)$$

Here i indexes the residential origin tract and j the workplace destination tract. The commuting count $\bar{N}_{ij}^{pre}$ is the average number of LODES workers traveling from i to j before the pandemic. The indicator A_{ijs} equals one when the modeled freeway route between the centroids of i and j traverses the freeway edge represented by PeMS station s . The set $\mathcal{S}_{c(i)}$ contains the stations in the county of origin tract i .

The index k runs over every candidate residential origin tract in $\mathcal{I}_{c(i)}$, including tract i . Thus the denominator is the total pre-pandemic routed-worker exposure to station s across all origin tracts in that county, and the ratio in brackets is the share assigned to tract i .

In the second bracket, $Flow_{sh}$ is observed traffic flow at station s in hour h , L_s is the freeway coverage length represented by that station in miles, and $\sum_{h \in t} Flow_{sh} L_s$ is annual station VMT. The allocation therefore does not weight every commuting pair equally. Routes that use more of the observed freeway network encounter more station exposure, while each station contributes only the VMT associated with its own coverage length. The pre-period route weights are fixed, so the pandemic movement in $\widehat{VMT}_{it}$ comes from observed PeMS traffic rather than from contemporaneous LODES commuting counts.

For every matched station, the bracketed allocation shares sum to one across origin tracts. Consequently,

$$\sum_{i \in c} \widehat{VMT}_{it} = \sum_{s \in c} VMT_{st}^{PeMS}. \quad (25)$$

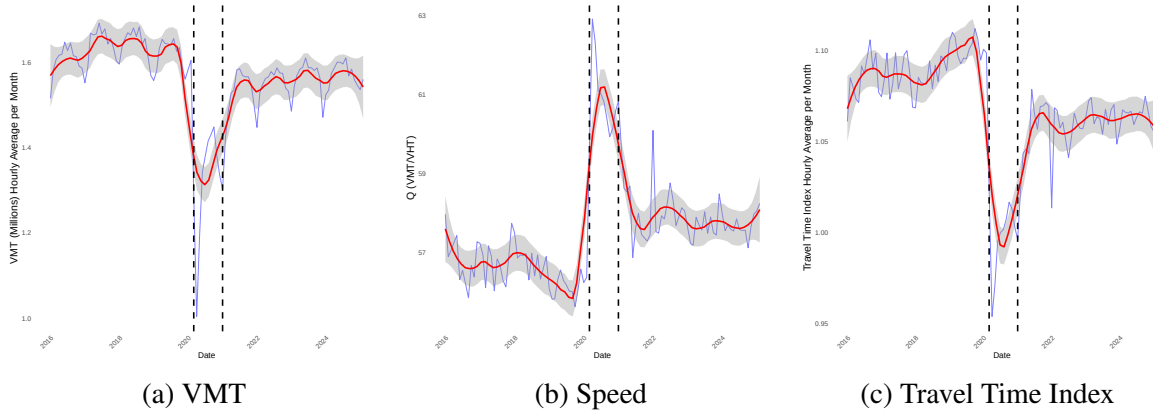
Stations without a route match are allocated using fixed baseline origin-worker shares, so the county accounting identity is preserved. Equation (25) shows that the allocation preserves observed county totals; it does not independently validate the tract-level estimates. The cross-sectional allocation comes from LODES and modeled route exposure; the aggregate time-series movement comes from observed PeMS traffic.

3.3 Remote work, industry exposure, and wages

ACS five-year estimates provide tract population, work-from-home shares, unemployment, median household income, poverty, college attainment, vehicle availability, civilian employment by industry, and earnings. LODES provides the origin–destination commuting counts used in Equation (24). Section 4 introduces the shift–share design after presenting the demand equation and explains how the industry variables identify differential exposure to national changes in remote work.

Figure 3 summarizes monthly traffic conditions across the five counties. Vehicle miles traveled (VMT) fell sharply in 2020, average speed rose, and the travel time index (TTI) fell. TTI is the ratio of observed travel time to travel time at the free-flow reference speed: a value of 1.2 indicates a trip taking 20 percent longer than the reference time. Traffic subsequently recovered in part, but the persistent change in commuting behavior motivates separating the pre-period 2018–2019 from the post-period 2021–2022.

Figure 3. Freeway Traffic and Congestion



Notes: Monthly summaries of hourly PeMS observations across the five counties. Panel (a) reports mean hourly county vehicle miles traveled (VMT), in millions. Panel (b) reports VMT-weighted speed in miles per hour; hourly $Q = VMT/VHT$, where VHT denotes vehicle hours traveled. Panel (c) reports the VMT-weighted travel time index (TTI), the ratio of observed travel time to the free-flow reference time; higher values indicate slower travel relative to that benchmark. Blue lines show monthly values and red lines show LOESS smooths. Dashed lines mark March 2020 and January 2021.

I summarize the persistent break in traffic conditions using adjusted before–after regressions that compare 2018–2019 with 2021–2022 and exclude 2020. Each regression relates log VMT, average speed, or TTI to a post-period indicator while absorbing county-by-hour-of-week and county-by-calendar-month fixed effects; standard errors are clustered by year-month. Appendix Table 7 shows that, across all hours, VMT was 7.71 percent lower, average speed was 3.08 percent higher, and TTI was 3.11 percent lower in the post period. During the weekday afternoon peak, the reduction in TTI was substantially larger, at 7.44 percent. These comparisons describe the change in traffic conditions; they are not the causal estimates of the effect of remote work developed in Section 4.

Appendix Figure 11 reports the adjusted TTI reduction separately by county. The decline ranges from 1.09 percent in Orange County to 5.25 percent in San Francisco, with Santa Clara also experiencing a relatively large reduction of 4.39 percent. This geographic heterogeneity motivates the county-specific congestion technologies used below. Appendix Figure 12 shows the corresponding increase in county work-from-home shares, while Ap-

pendix Figure 13 documents how the changes in VMT and congestion vary over the 24-hour day. Together with Figure 3, these results show that the pandemic shift affected the level, timing, and geography of freeway use.

4 Estimation

4.1 Congestion technology

Let Q_{cht} denote observed average speed. Proportional delay relative to free flow is

$$D_{cht} = \max \left\{ \frac{S_c^0}{Q_{cht}} - 1, 0 \right\}. \quad (26)$$

The truncation at zero treats speeds above the estimated free-flow benchmark as no measured congestion rather than negative delay. The preferred conditional-mean specification is

$$E[D_{cht} \mid F_{cht}, c] = \alpha_c \left(\frac{F_{cht}}{K_c} \right)^\beta, \quad (27)$$

estimated by Poisson pseudo-maximum likelihood (PPML), following the conditional-mean approach discussed by [Santos Silva and Tenreyro \(2006\)](#). PPML is useful here because D_{cht} is nonnegative and frequently zero; log OLS would discard zero-delay observations and mechanically select the congested part of the sample. The estimator is used as a quasi-likelihood for the conditional mean and does not require D_{cht} to be a count.

The preferred specification contains county-specific α_c and one common β . It uses 175,180 county-hour observations in 2018, 2019, 2021, and 2022. The estimated curvature is $\hat{\beta} = 3.270$ with an HC1 standard error of 0.019. Table 1 reports the county scale estimates. Because flow and speed are jointly determined within the hour, I interpret Equation (27) as the congestion curve used for structural calibration rather than as a fully causal

supply estimate.

4.2 Tract-level demand

The empirical demand equation is

$$\ln \widehat{VMT}_{it} = \xi_i + \delta_{c(i)t} + \rho WFH_{it} + \gamma \ln Population_{it} + X'_{it}\pi + \varepsilon_{it}, \quad (28)$$

where WFH_{it} is measured in percentage points, ξ_i are tract fixed effects, and $\delta_{c(i)t}$ are county-by-year fixed effects. County-by-year effects absorb shocks to traffic common to tracts in the same county and year, including the common effects of fuel and other operating costs, county labor-market conditions, and policies. The preferred specification sets X_{it} empty; a robustness specification adds unemployment, household income, poverty, college attainment, and vehicle availability.

Because remote work is endogenous, I instrument it with

$$Z_{it} = 100 \sum_q s_{iq,2019} \left(WFH_{q,t}^{US \setminus CA} - WFH_{q,2019}^{US \setminus CA} \right), \quad (29)$$

where $s_{iq,2019}$ is the predetermined employment share of industry q in tract i , and the term in parentheses is the national change in that industry's work-from-home rate calculated outside California. The identifying assumption is that, conditional on tract and county-by-year fixed effects, predetermined differences in local industry exposure affect tract freeway VMT through their effect on remote work rather than through a separate tract-specific channel aligned with the national industry shock.

Equation (29) is an exposure design in the terminology of [Goldsmith-Pinkham et al. \(2020\)](#) and a shock-based shift-share design in the terminology of [Borusyak et al. \(2022\)](#). The empirical argument relies on the national industry shocks being as-good-as-random with

respect to the appropriately weighted tract residuals, while the 2019 shares determine differential exposure.

Table 2 presents the estimates. The preferred coefficient is -0.001441 , so a 10 percentage point increase in remote work changes VMT by $100[\exp(10\hat{\rho}) - 1] = -1.43$ percent. Adding demographic controls gives a similar effect of -1.16 percent. The clustered first-stage Wald statistics are large, and standard errors are two-way clustered by tract and year.

Table 2. Remote Work and Tract-Level Freeway VMT

	(1) Preferred IV	(2) IV with controls
Work-from-home share (percentage points)	-0.001441^{**} (0.000582)	-0.001167^* (0.000486)
Effect of a 10 p.p. increase (percent)	-1.43	-1.16
First-stage coefficient	3.606	3.622
First-stage clustered Wald statistic	1,006.6	744.9
Observations	25,088	24,975
Tract fixed effects	Yes	Yes
County-by-year fixed effects	Yes	Yes
Log population	Yes	Yes
Additional demographic controls	No	Yes

Notes: The dependent variable is log tract-level freeway VMT. The work-from-home share is instrumented with 2019 tract industry shares interacted with national, leave-California-out changes in industry-specific work-from-home rates. Column (2) additionally controls for unemployment, median household income, poverty, college attainment, and the share of households without a vehicle. Standard errors in parentheses are two-way clustered by tract and year. The first-stage statistic is the clustered Wald statistic for the excluded instrument. Because the panel contains seven years, the reported inference should be interpreted cautiously; exposure-robust shift-share inference is an important next step. $^*p < 0.10$, $^{**}p < 0.05$.

The estimated remote-work and population responses enter the demand shifters used in the hourly congestion-pricing simulations.

5 Results

5.1 Optimal taxes with fixed and contemporaneous values of time

Table 3 reports average optimal taxes. Under a fixed 2018 value of time, post-period taxes fall in every county. The reductions range from 2.39 percent in San Diego to 4.75 percent in San Francisco. These estimates isolate the physical effect of the demand shift: lower efficient traffic reduces marginal external delay.

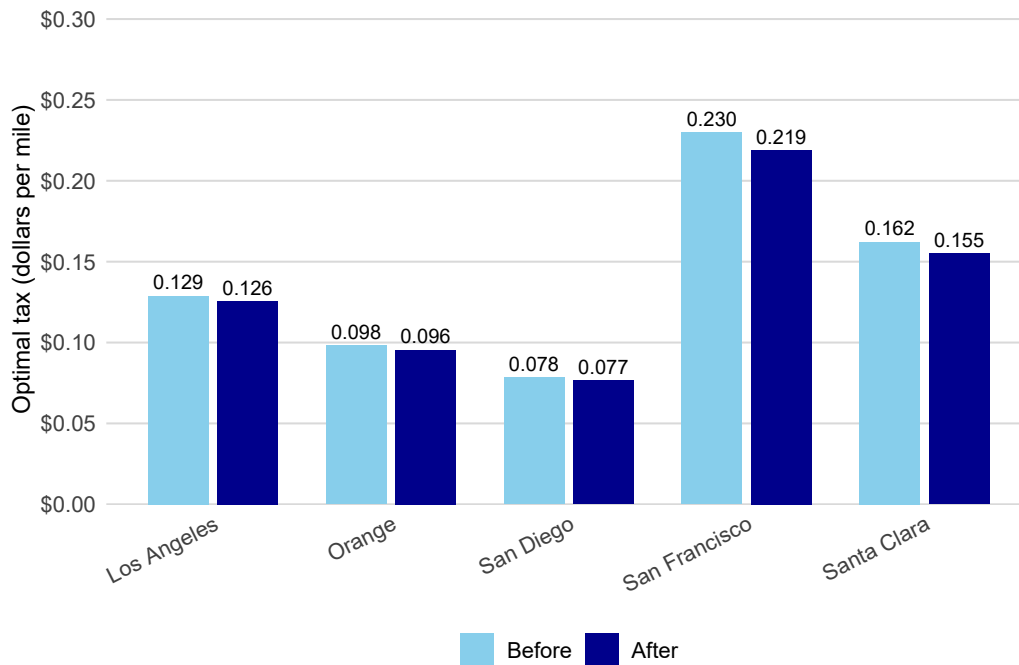
Table 3. Optimal Congestion Taxes Before and After the Demand Shift

County	Fixed pre-period value of time			Contemporaneous value of time		
	Pre	Post	Change (%)	Pre	Post	Change (%)
Los Angeles	0.129	0.126	−2.52	0.130	0.145	11.38
Orange	0.098	0.096	−2.78	0.098	0.109	10.84
San Diego	0.078	0.077	−2.39	0.079	0.088	10.95
San Francisco	0.230	0.219	−4.75	0.231	0.250	8.30
Santa Clara	0.162	0.155	−4.42	0.165	0.178	7.98

Notes: Taxes are average rates in dollars per vehicle mile over the county’s hourly profile. The pre-period is 2018–2019 and the post-period is 2021–2022. The fixed-value-of-time calculation holds the county value of time at its 2018 level. The contemporaneous calculation uses one-half of the county median hourly wage in each year. All equilibria impose a private-cost elasticity of -0.1 .

Figure 4 shows the same counterfactual. The levels differ across counties because the estimated congestion curves, network capacities, baseline hourly profiles, and wages differ. Changes in work-from-home shares and population also vary across tracts and counties, producing different county demand shifts and therefore different post-period traffic and taxes. San Francisco has the largest dollar-per-mile tax even though its observed detector network is much smaller, reflecting its steep congestion technology and high value of time.

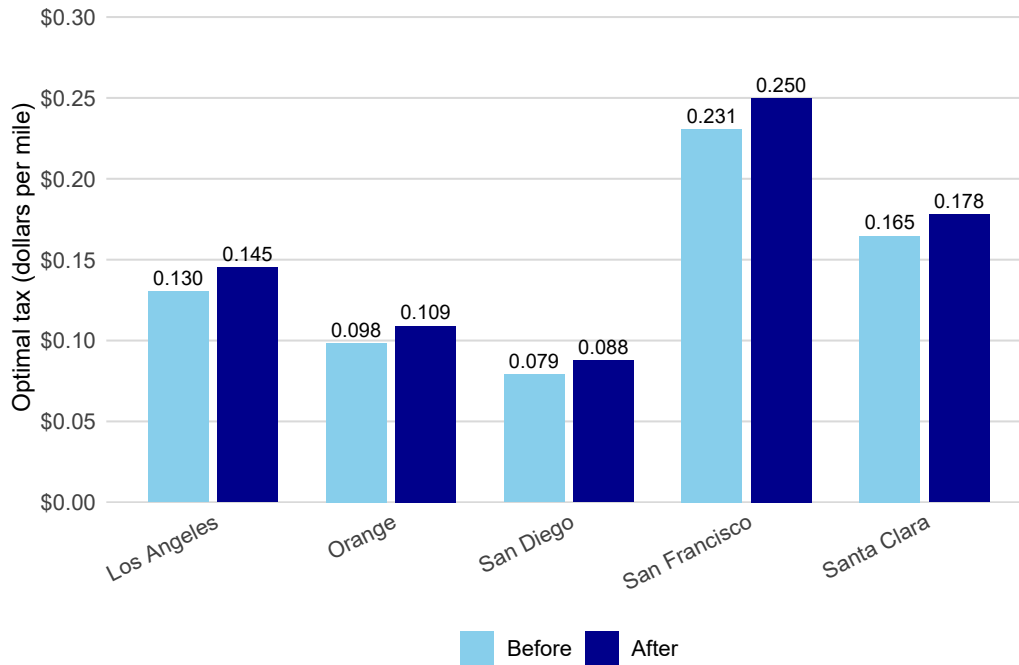
Figure 4. Optimal Congestion Taxes with the Value of Time Fixed



Notes: Average county-period taxes in dollars per mile. The value of time is fixed at its 2018 county level.

Allowing wages to change reverses the comparison. As shown in Figure 5, post-period taxes rise by 7.98 percent in Santa Clara, 8.30 percent in San Francisco, and approximately 11 percent in Los Angeles, Orange, and San Diego. The fixed-wage calculation understates the contemporaneous post-period tax by 12.4–13.4 percent.

Figure 5. Optimal Congestion Taxes with Contemporaneous Values of Time



Notes: Average county-period taxes in dollars per mile. The value of time equals one-half of the contemporaneous county median hourly wage.

5.2 Why the tax rises

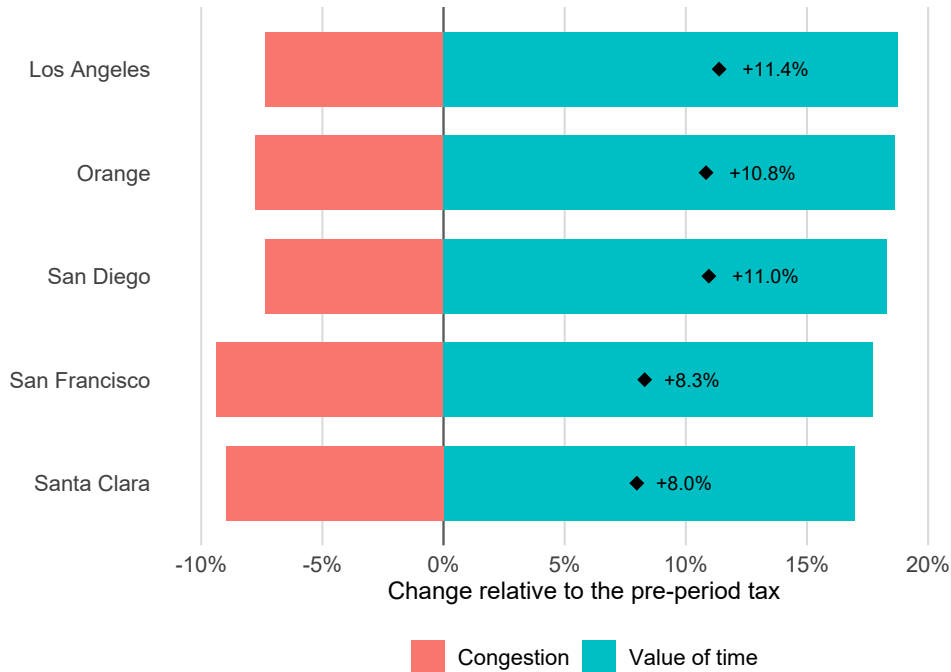
Table 4 and Figure 6 apply Equation (22). The congestion component is negative in every county, ranging from -7.35 to -9.40 percent of the pre-period tax. The value-of-time component is positive and substantially larger, from 16.96 to 18.76 percent. The net effect is therefore an increase in the optimal monetary charge.

Table 4. Decomposition of the Change in the Optimal Tax

County	Congestion component (% of pre tax)	Value-of-time component (% of pre tax)	Total change (% of pre tax)
Los Angeles	−7.38	18.76	11.38
Orange	−7.80	18.63	10.84
San Diego	−7.35	18.30	10.95
San Francisco	−9.40	17.71	8.30
Santa Clara	−8.99	16.96	7.98

Notes: The exact decomposition writes $\tau = VOT \times g$, where g is marginal external delay at the efficient flow. The congestion component holds the value of time at its pre-period level while moving from g_{pre} to g_{post} . The value-of-time component prices post-period delay using the change in the value of time. Components add exactly to the total percentage change, up to rounding.

Figure 6. Decomposition of the Change in the Optimal Tax



Notes: Components are expressed as percentages of the pre-period contemporaneous-wage tax. The congestion and value-of-time components add exactly to the total change.

This result should not be interpreted as evidence that the pandemic increased physical congestion. It did not. Rather, the dollar value attached to a unit of marginal delay rose by

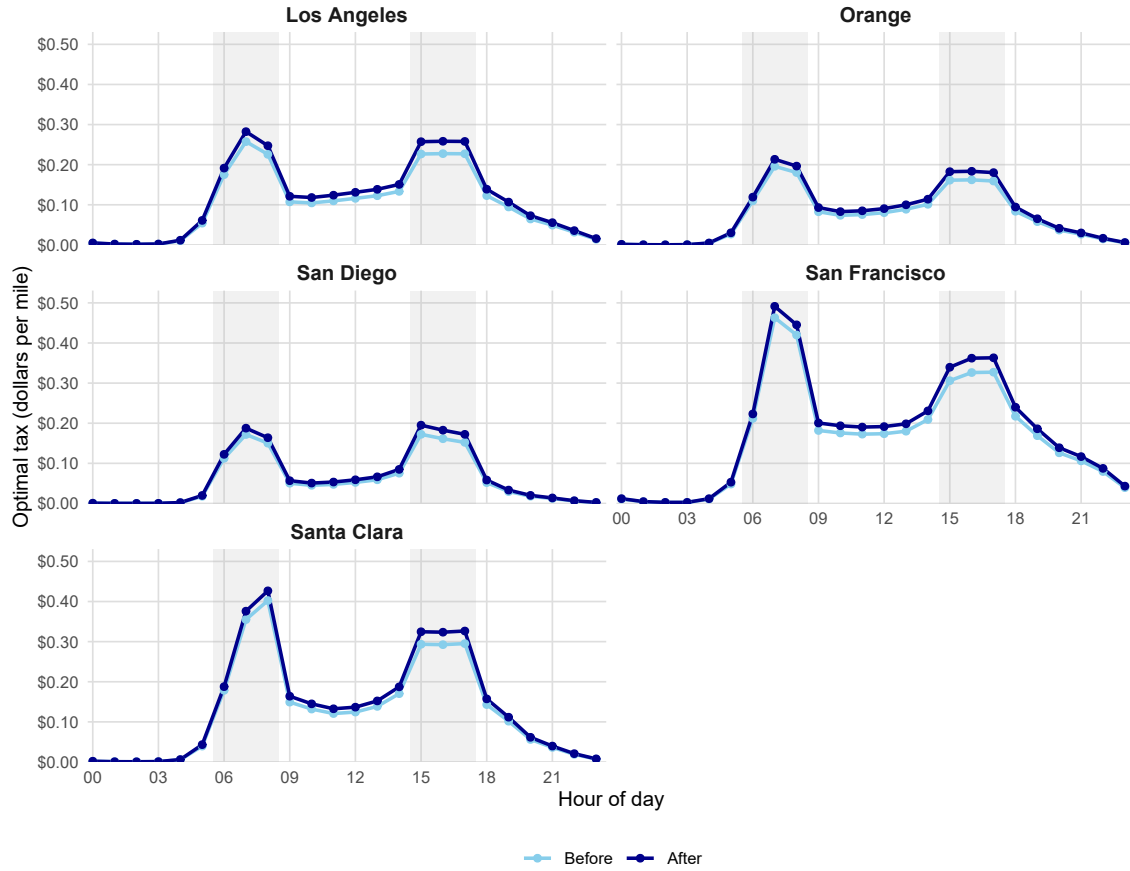
more than the marginal delay itself fell. The sign difference between the fixed- and changing-wage exercises is exactly the distinction between physical congestion and its monetary social cost.

5.3 Time of day

The hourly profile is economically important because the marginal external cost is highly nonlinear in flow. I re-estimate Equation (27) separately for weekday morning peak hours (6:00–8:59), afternoon peak hours (15:00–17:59), and all other weekday hours. Within each period, the tax is still solved for every hour rather than assigned as a single period-wide rate.

Figure 7 plots the contemporaneous-wage hourly schedule. Taxes are nonnegative in every county and period: several off-peak hourly rates round to zero, and the model never predicts a negative tax or subsidy. The largest rates occur around the commuting peaks. San Francisco and Santa Clara have especially pronounced morning profiles, while Los Angeles and San Diego exhibit relatively large afternoon rates. The post-period schedule is generally above the pre-period schedule because higher values of time dominate lower marginal delay.

Figure 7. Optimal Congestion Tax over the Weekday Hourly Profile



Notes: County-hour optimal taxes use contemporaneous values of time. Shaded bands denote 6:00–8:59 a.m. and 3:00–5:59 p.m. Morning, afternoon, and off-peak congestion technologies are estimated separately; efficient flow and tax are solved for each hour within those periods.

The period averages in Appendix Figure 9 range from roughly 4 to 14 cents per mile off peak in the pre-period and from 15 to 37 cents per mile during peak periods. With contemporaneous wages, post-period changes are positive in all 15 county–time-of-day combinations (five counties and three times of day), ranging from 5.85 to 13.54 percent.

5.4 Income heterogeneity and group-specific tax rates

I divide tracts into groups $g \in \{L, H\}$. A tract belongs to the higher-income group when its 2018 median household income is above the household-weighted 80th percentile in its

county; the remaining tracts form the lower-income group. This classification is held fixed over time. Let $\tilde{w}_{gct}$ be the employed-person-weighted median of tract median annual earnings within county-income group gc , divided by 2,080 hours.² The group value of time is

$$V_{gct} = \theta \tilde{w}_{gct}, \quad \theta = 0.5. \quad (30)$$

Thus V_{gct} is group g 's value of travel time in dollars per hour in county c and year t ; it is one-half of the group's estimated hourly wage, not its annual household income.

The computation begins by estimating the tract demand equation separately for each group:

$$\ln \widehat{VMT}_{it} = b_g WFH_{it} + \rho_g \ln Population_{it} + \mu_i + \delta_{c(i)t} + e_{it}, \quad i \in \mathcal{I}_{gc}. \quad (31)$$

As in the pooled specification, WFH_{it} is instrumented with the leave-California-out shift-share measure. Let A_{gcht} denote the demand shifter for income group g in county c , weekly hour h , and year t . It incorporates the group's baseline traffic and its estimated demand changes associated with remote work and population. Demand can therefore evolve differently across groups because both their estimated responses and their observed changes in remote work and population differ. Congestion depends on the groups' combined use of the road, $F_{cht} = F_{Lcht} + F_{Hcht}$.

As in the heterogeneous-user congestion-pricing framework of [Verhoef and Small \(2004\)](#), marginal external cost accounts for the values of time of all travelers exposed to the added delay. In the shared-network model here, this implies a common monetary Pigouvian toll. The superscript P denotes the unconstrained Pigouvian allocation; it is a policy label, not the generalized price p used in Equation (9). Thus F_{gcht}^P is efficient flow supplied by group g ,

²The conversion assumes 40 hours of work per week for 52 weeks, giving $40 \times 52 = 2,080$ hours per year. This is a common annual-hours assumption used to convert earnings into an hourly wage, not a measure of each worker's actual hours.

F_{cht}^P is their total, and τ_{cht}^P is the common first-best monetary toll in dollars per mile. These objects solve

$$F_{gcht}^P = A_{gcht} [V_{gct} T_c(F_{cht}^P) + \tau_{cht}^P]^{-\eta}, \quad F_{cht}^P = \sum_g F_{gcht}^P, \quad (32)$$

$$\tau_{cht}^P = T'_c(F_{cht}^P) \sum_g V_{gct} F_{gcht}^P. \quad (33)$$

Every additional vehicle mile creates the same physical delay, regardless of which group supplies it, so Equation (33) is the same dollar-per-mile charge for both groups. The additive terms $\tau_{gcht}^{contrib} = T'_c(F_{cht}^P) V_{gct} F_{gcht}^P$ sum to τ_{cht}^P and attribute the external cost to the travelers who bear the added delay. They are contributions to the common toll, not rates levied on each group. Appendix Figure 10 reports this decomposition.

[Pandey et al. \(2022, Proposition 1\)](#) provide a theoretical precedent for a common time-equivalent charge. Their deterministic express-lane model uses discounts that imply a common toll-to-value-of-time ratio across traveler groups. To construct group-specific rates in this paper, I impose the same proportionality restriction, so both groups face the same toll measured in travel time, κ_{cht} hours per mile, converted into money using the group value of time,

$$\tau_{gcht}^V = V_{gct} \kappa_{cht}. \quad (34)$$

Here κ_{cht} is the common time-equivalent charge: it adds the same number of travel hours per mile to each group's perceived time cost. It is not itself a dollar toll. Multiplication by V_{gct} converts it into the group-specific monetary rate τ_{gcht}^V , where the superscript V denotes the value-of-time-proportional policy. The generalized monetary price is therefore $V_{gct} [T_c(F_{cht}) + \kappa_{cht}]$.

[Pandey et al. \(2022\)](#) examine this rule in express-lane simulations. To my knowledge, this paper is the first to use empirical estimates of congestion technology and income-specific

remote-work demand shifts to compute the optimal common time-equivalent congestion charge. Computing the common time-equivalent charge itself is an empirical contribution of the paper, together with calculating the corresponding monetary tolls for each income group and examining how both change with remote work and wages.

With the common elasticity η , the constrained optimum and the two group flows are the joint solution to

$$F_{gcht}^\kappa = A_{gcht} \{V_{gct} [T_c(F_{cht}^\kappa) + \kappa_{cht}^*]\}^{-\eta}, \quad F_{cht}^\kappa = \sum_g F_{gcht}^\kappa, \quad (35)$$

$$\kappa_{cht}^* = F_{cht}^\kappa T'_c(F_{cht}^\kappa). \quad (36)$$

The superscript κ on flow denotes equilibrium under this constrained schedule, while κ_{cht}^* is the welfare-maximizing common time charge within that policy class. Substituting the BPR derivative from Equation (2) makes the tax computed in each county-hour explicit:

$$\kappa_{cht}^* = \frac{\alpha_c \beta}{S_c^0} \left(\frac{F_{cht}^\kappa}{K_c} \right)^\beta, \quad (37)$$

$$\tau_{gcht}^V = \frac{V_{gct}}{S_c^0} \alpha_c \beta \left(\frac{F_{cht}^\kappa}{K_c} \right)^\beta. \quad (38)$$

Equations (35)–(38) are solved simultaneously, rather than assigning each group a separate road or holding total traffic fixed. Within the same county-hour, both groups face the same κ_{cht}^* , so

$$\frac{\tau_{Hcht}^V}{\tau_{Lcht}^V} = \frac{V_{Hct}}{V_{Lct}}. \quad (39)$$

This proportional schedule is not the unconstrained first-best common monetary toll in Equation (33); it is the optimum within the restricted class $\tau_g = V_g \kappa$. Appendix E derives both results from welfare.

Figure 8 and Table 5 report the constrained rates. Lower-income tracts account for ap-

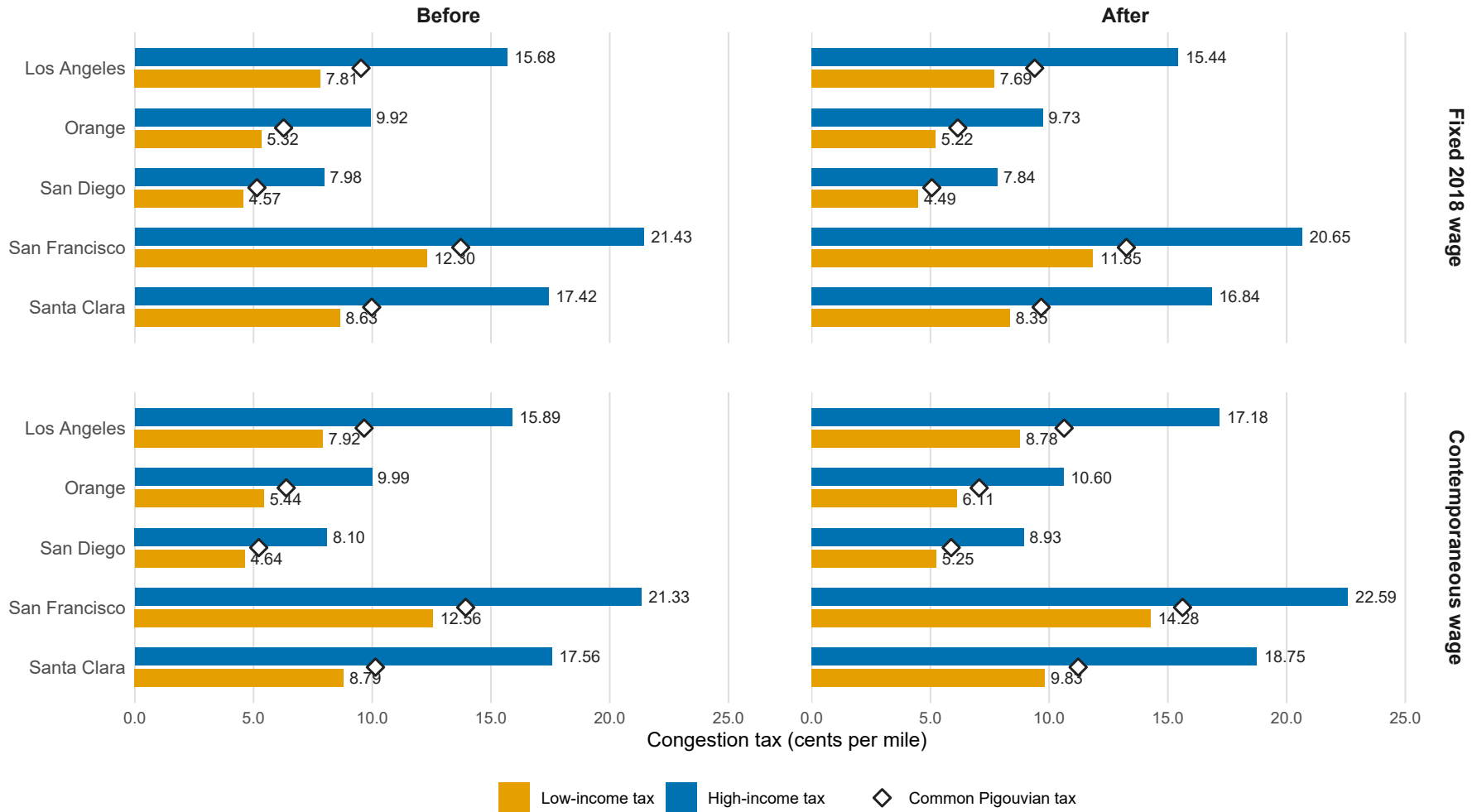
proximately 77–84 percent of equilibrium flow and therefore for most of the common Pigouvian externality. Nonetheless, their proportional rates are lower because Equation (39) maps the common time charge into dollars through group-specific values of time. In the post-period, higher-income rates are 1.6–2.0 times lower-income rates. For example, the Los Angeles rates are 17.2 and 8.8 cents per mile, respectively. With contemporaneous wages, the post-period rate rises in every county and for both groups: by 10.8–13.7 percent for lower-income tracts and 5.9–10.3 percent for higher-income tracts.

Table 5. Value-of-Time-Proportional Taxes by Income Group

County	Lower-income tracts			Higher-income tracts		
	Pre	Post	Change (%)	Pre	Post	Change (%)
Los Angeles	0.079	0.088	10.84	0.159	0.172	8.14
Orange	0.054	0.061	12.41	0.100	0.106	6.19
San Diego	0.046	0.052	13.17	0.081	0.089	10.25
San Francisco	0.126	0.143	13.73	0.213	0.226	5.91
Santa Clara	0.088	0.098	11.77	0.176	0.187	6.73

Notes: Average dollar-per-mile rates under the constrained policy $\tau_g = V_g \kappa$, where the common time-equivalent toll is $\kappa = F^\kappa T'(F^\kappa)$. Group demands and total congestion are solved jointly for each county, hour, and year. Higher-income tracts are above the household-weighted county 80th percentile of 2018 tract median household income; the classification is held fixed. Group values of time equal one-half of employed-person-weighted median hourly earnings within each county-income group. These are actual group-specific rates under the proportional restriction, not the groups' contributions to the unconstrained common Pigouvian toll.

Figure 8. Value-of-Time-Proportional Congestion Taxes by Income Group



Notes: Bars report the constrained group-specific monetary rates $\tau_g = VOT_g \kappa$; the rates and traffic are solved jointly on the shared county network. Diamonds show the common monetary Pigouvian benchmark. The upper panels hold wages at 2018 levels, and the lower panels use contemporaneous wages. Before denotes 2018–2019 and after denotes 2021–2022.

6 Conclusion

The paper’s main contribution is an empirical congestion-pricing framework that combines income-group differences in remote-work exposure, estimated demand responses, traffic shares, and the value of time. I estimate remote-work responses separately for lower- and higher-income tracts and use their observed changes in remote work and population to construct group-specific demand shifts, rather than imposing the same shock on all travelers. Both groups’ demands and congestion are then solved jointly on the shared freeway network.

The shift–share demand estimates imply that a 10 percentage point increase in the share working from home reduces allocated tract freeway VMT by 1.43 percent. Under the design’s identifying assumptions, this is a causal effect of remote work on traffic demand. The group-specific estimates carry differences in remote-work exposure and behavior into the pricing exercise.

The pricing results show that less congestion can coexist with higher optimal monetary tolls. Holding the value of time at its 2018 level, taxes fall by 2.4–4.7 percent after the demand shift. Using contemporaneous wages, they instead rise by 8.0–11.4 percent. The decomposition shows that the increase in the dollar value of time outweighs the decline in marginal physical delay. The heterogeneous-agent exercise further shows that lower-income tracts supply most modeled traffic, while higher-income travelers face monetary rates 1.6–2.0 times larger under the constrained common time-equivalent schedule. These rate comparisons are distinct from net welfare effects by income group.

Heterogeneity matters because a demand shock to one group changes travel time for everyone on the shared network. The framework captures both the group’s own demand adjustment and the resulting change in delay borne by other travelers. It therefore connects income-specific remote-work responses to traffic composition and congestion prices, rather than representing the demand shift through a single average traveler.

The estimated tax rates depend on the data and assumptions used in the analysis and should be interpreted as policy benchmarks rather than precise prescriptions for implementation. The paper provides a framework that policymakers can use when setting congestion taxes to account for changes in traffic demand, the value of time, and differences across travelers.

For example, the heterogeneous-agent exercise illustrates an implicit cross-subsidy under the common time-equivalent charge. Evaluated at the same traffic allocation, higher-income travelers pay more per mile than the common marginal external congestion cost, while lower-income travelers pay less. The higher-income group's payments above this benchmark cover the gap left by the lower-income group's payments, allowing lower-income traffic to face a lower monetary rate. This interpretation concerns the sharing of congestion charges across groups.

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A Derivation of Private, External, and Social Costs

The distinction between private and social congestion costs follows the classical analysis of [Walters \(1961\)](#) and the textbook treatment in [Small et al. \(2024, Chapter 3\)](#). The derivation below applies this framework to the travel-time costs in the paper's model.

This appendix suppresses the county, hour, and year indices for readability. Let F denote modeled freeway flow in passenger-car-equivalent vehicle miles per hour, $T(F)$ travel time in hours per mile, and V the value of time in dollars per hour. Thus V corresponds to VOT_{ct} in the main text. A prime denotes differentiation with respect to flow. The abbreviations PMC , $MECC$, and MSC mean private marginal cost, marginal external congestion cost, and marginal social cost, respectively, all in dollars per mile. Aggregate travel-time cost $C(F)$ is

$$C(F) = V \times F \times T(F), \quad (40)$$

measured in dollars per hour. A nonatomic driver treats $T(F)$ as given. Her private cost of one additional vehicle mile is therefore

$$PMC(F) = V \times T(F). \quad (41)$$

The planner recognizes that an additional unit of flow changes the travel time experienced by every unit already on the road:

$$MSC(F) = \frac{dC(F)}{dF} \quad (42)$$

$$= V \times [T(F) + F \times T'(F)] \quad (43)$$

$$= \underbrace{V \times T(F)}_{PMC(F)} + \underbrace{V \times F \times T'(F)}_{MECC(F)}. \quad (44)$$

This is the origin of Equations (6) and (7): the first term is the entrant's own time cost and

the second is the time cost imposed on all existing flow.

For the capacity-normalized Bureau of Public Roads (BPR) function ([U.S. Bureau of Public Roads, 1964](#)), let S^0 be free-flow speed in miles per hour, K network capacity in the same units as F , $\alpha > 0$ the congestion scale parameter, and $\beta > 0$ the curvature parameter. Travel time is

$$T(F) = \frac{1}{S^0} \times [1 + \alpha \times (F/K)^\beta], \quad (45)$$

the derivative is

$$T'(F) = \frac{\alpha \times \beta}{S^0 \times K} \times (F/K)^{\beta-1}. \quad (46)$$

Multiplying by F gives

$$F \times T'(F) = \frac{\alpha \times \beta}{S^0} \times (F/K)^\beta. \quad (47)$$

Therefore,

$$PMC(F) = \frac{V}{S^0} \times [1 + \alpha \times (F/K)^\beta], \quad (48)$$

$$MECC(F) = \frac{V}{S^0} \times \alpha \times \beta \times (F/K)^\beta, \quad (49)$$

$$MSC(F) = \frac{V}{S^0} \times [1 + \alpha \times (1 + \beta) \times (F/K)^\beta]. \quad (50)$$

All three expressions are dollars per mile. Capacity is essential for interpreting the flow ratio, while free-flow speed determines the conversion from a time index to hours per mile.

B Efficient Flow, Optimal Toll, and Pass-Through

The welfare argument follows the standard static congestion-pricing model presented in [Small et al. \(2024, Section 4.1\)](#). Social surplus is the benefit from travel net of its resource costs. Toll payments are transfers from travelers to the toll recipient, so they are not deducted again as a social cost.

This appendix suppresses the county, hour, and year indices for readability. Let F denote modeled traffic flow, $T(F)$ travel time in hours per mile, and V the value of time in dollars per hour. Let $P(F)$ denote inverse demand, or marginal willingness to pay for freeway travel, in dollars per mile.³ Write A for the demand shifter, η for the common price-elasticity magnitude, and τ for the monetary toll per mile. Private marginal cost is $PMC(F) = V \times T(F)$, marginal external congestion cost is $MECC(F) = V \times F \times T'(F)$, and marginal social cost is $MSC(F) = PMC(F) + MECC(F)$. A prime denotes a flow derivative and a star an efficient value.

Let $F^{ref} > 0$ be a fixed reference flow. Net social surplus $W(F)$, up to an additive constant, is

$$W(F) = \int_{F^{ref}}^F P(x) dx - V \times F \times T(F). \quad (51)$$

Here x is the variable of integration used to add the marginal benefits of successive units of travel. All travelers experience the travel time at total flow F , so total time cost is $V \times F \times T(F)$. A positive reference makes welfare finite under the calibrated constant-elasticity demand, whose integral from zero diverges when $\eta \leq 1$. The reference changes neither welfare comparisons nor the optimality condition.

To find efficient flow, differentiate Equation (51) with respect to F , holding V and F^{ref} fixed. The fundamental theorem of calculus gives

$$\frac{d}{dF} \left[\int_{F^{ref}}^F P(x) dx \right] = P(F).$$

For total travel-time cost, both the amount of travel F and the time per mile $T(F)$ change

³In the main text's notation, $P_{cht}(F) = (A_{cht}/F)^{1/\eta}$ is the inverse of $F_{cht} = A_{cht}p_{cht}^{-\eta}$. Here A_{cht} is the demand shifter and $\eta = 0.1$ is the absolute value of the price elasticity. At a private equilibrium, $P_{cht}(F_{cht}) = p_{cht} = PMC_{cht}(F_{cht}) + \tau_{cht}$. The function $P(F)$ therefore denotes willingness to pay, whereas p_{cht} denotes the generalized price faced by drivers.

with flow. The product rule gives

$$\frac{d}{dF} [V \times F \times T(F)] = V \times \left[\frac{dF}{dF} \times T(F) + F \times \frac{dT(F)}{dF} \right] = V \times T(F) + V \times F \times T'(F).$$

Subtracting the cost derivative from the benefit derivative yields

$$W'(F) = P(F) - V \times T(F) - V \times F \times T'(F).$$

The planner sets this derivative equal to zero at the optimal flow F^* and rearranges the terms, giving

$$P(F^*) = V \times [T(F^*) + F^* \times T'(F^*)] = MSC(F^*). \quad (52)$$

Private equilibrium under a tax τ satisfies $P(F) = V \times T(F) + \tau$. Evaluating this condition at F^* and comparing it with Equation (52) proves that

$$\tau^* = V \times F^* \times T'(F^*) = MECC(F^*). \quad (53)$$

For the constant-elasticity demand $F = A \times [PMC(F) + \tau]^{-\eta}$, log differentiation yields

$$\frac{1}{F} \times \frac{dF}{d\tau} = -\eta \times \frac{PMC'(F) \times (dF/d\tau) + 1}{PMC(F) + \tau}, \quad (54)$$

which rearranges to Equation (15). Because $PMC'(F) = V \times T'(F)$, the optimal tax also satisfies $\tau^* = F^* \times PMC'(F^*)$. Substitution into Equation (17) gives

$$\begin{aligned} \left. \frac{dp}{d\tau} \right|_{\tau=\tau^*} &= \frac{PMC(F^*) + MECC(F^*)}{PMC(F^*) + MECC(F^*) + \eta \times MECC(F^*)} \\ &= \frac{MSC(F^*)}{MSC(F^*) + \eta \times MECC(F^*)}. \end{aligned} \quad (55)$$

Here $p = PMC(F) + \tau$ is the generalized private price in dollars per mile, so $dp/d\tau$ mea-

asures how much of a marginal toll increase reaches the driver after the resulting congestion relief. The expression lies strictly between zero and one whenever congestion is present. It characterizes the marginal incidence of the optimal toll; it is not a separate rule for selecting its level.

C Tract Freeway VMT and the Shift–Share Design

In this appendix, i and k index residential origin tracts, j workplace destination tracts, s freeway detector stations, h observed hours, and t calendar years. The county containing tract i is $c(i)$; $\mathcal{I}_{c(i)}$ is its set of candidate origin tracts and $\mathcal{S}_{c(i)}$ its observed stations. The outcome $\widehat{VMT}_{it}$ is annual vehicle miles traveled allocated to tract i , not a direct count of that tract’s trips. The pre-period commuting count $\overline{N}_{ij}^{pre}$ comes from the Longitudinal Employer-Household Dynamics Origin-Destination Employment Statistics (LODES). The route indicator A_{ijs} equals one when the modeled route from i to j passes station s ; it is not the demand shifter A_{cht} . $Flow_{sh}$ denotes observed vehicles per hour and L_s station coverage length in miles. Station observations come from California’s Performance Measurement System (PeMS).

Equation (24) combines two distinct sources of variation. The numerator $\sum_j \overline{N}_{ij}^{pre} A_{ijs}$ measures the pre-pandemic commuting exposure of origin tract i to station s . The denominator repeats the same calculation over every candidate origin k in the county. For any matched station, summing that ratio over i therefore gives one. The second bracket, $\sum_{h \in t} Flow_{sh} L_s$, is the station’s observed annual VMT contribution. Summing Equation (24) over origin tracts consequently reproduces total PeMS VMT for the county, as stated in Equation (25).

The indices i and j uniquely identify an origin–destination pair; no redundant county index is needed inside the commuting-pair sums. County enters only through the origin-tract set $\mathcal{I}_{c(i)}$ and the station set $\mathcal{S}_{c(i)}$. Because the LODES counts and route indicators are

fixed in the pre-period, they determine only how observed station VMT is distributed across tracts. Year-to-year traffic movement comes from PeMS. The resulting $\widehat{VMT}_{it}$ should therefore be interpreted as exposure-allocated freeway VMT associated with tract i ; changes in unobserved route choice within an origin tract are not measured.

For the first-stage regression, WFH_{it} is the tract's work-from-home share in percentage points, $Population_{it}$ its population, and Z_{it} the shift-share instrument combining 2019 tract industry employment shares with national industry-specific changes in remote work calculated outside California. The first stage is

$$WFH_{it} = \xi_i^F + \delta_{c(i)t}^F + \pi Z_{it} + \gamma_F \ln Population_{it} + X_{it}' \lambda + u_{it}. \quad (56)$$

Here ξ_i^F and $\delta_{c(i)t}^F$ are tract and county-by-year fixed effects in the first stage; the superscript F labels that stage, not traffic flow. The coefficients π and γ_F capture the instrument and log-population relationships. X_{it} is the optional vector of demographic controls, λ its coefficient vector, and u_{it} the regression error; the prime on X_{it} denotes transpose. The leave-California-out construction limits mechanical correlation between local remote-work measurement and the national industry shocks.

D Time-of-Day Congestion Technology

Let c index counties, h observed hours, and t calendar years. F_{cht} is traffic flow in passenger-car-equivalent vehicle miles per hour, Q_{cht} is observed average speed in miles per hour, S_c^0 is county free-flow speed in miles per hour, and K_c is network capacity in the same units as F_{cht} . Proportional delay above free flow is

$$D_{cht} = \max \{ S_c^0 / Q_{cht} - 1, 0 \}.$$

Thus $D_{cht} = 0.2$ means travel time per mile is 20 percent above the free-flow benchmark, while speeds above that benchmark are assigned zero delay. This is a dimensionless delay measure, not travel time in hours. Period r denotes morning peak (6:00–8:59), afternoon peak (15:00–17:59), or all other weekday hours. I estimate

$$E[D_{cht} \mid F_{cht}, c, r] = \alpha_{cr} \left(\frac{F_{cht}}{K_c} \right)^{\beta_r}. \quad (57)$$

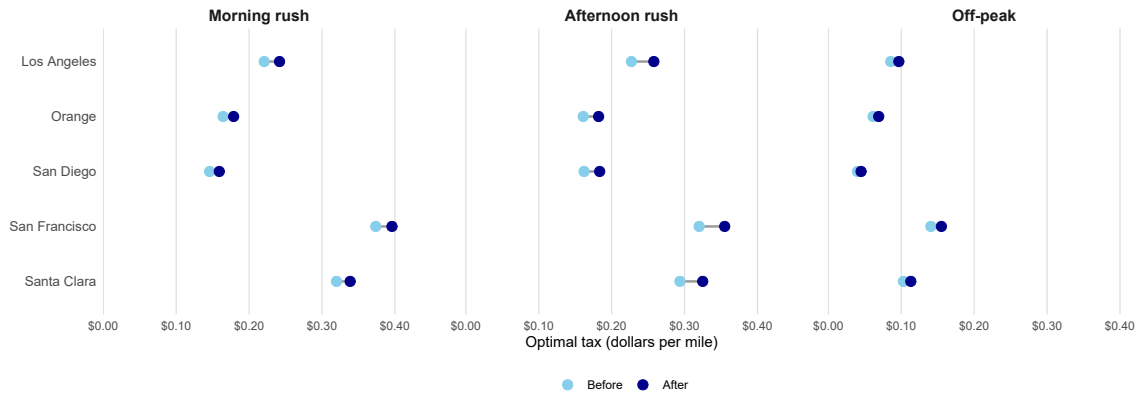
Here $E[\cdot \mid \cdot]$ is a conditional mean, $\alpha_{cr} > 0$ is a county-period congestion scale, and $\beta_r > 0$ is a curvature parameter common to counties within period r . Each period’s Bureau of Public Roads (BPR) curve is fitted by Poisson pseudo-maximum likelihood (PPML). Capacity and free-flow speed remain county-specific engineering inputs, while the BPR scale and curvature can differ across periods. This allows the same flow-capacity ratio to produce different expected delay at different times of day, reflecting systematic changes in bottlenecks and network conditions.

Table 6. Congestion Curvature by Time of Day

Period	Hours	$\hat{\beta}$	HC1 standard error
Morning peak	6:00–8:59	4.240	0.040
Afternoon peak	15:00–17:59	2.394	0.089
Off-peak	Other weekday hours	2.538	0.021

Notes: Each row comes from a separate weekday PPML BPR regression with a common curvature parameter within the period and county-specific scale parameters. The morning and afternoon regressions each use 15,645 observations; the off-peak regression uses 93,870. The period-specific congestion curve is then used to solve the hourly efficient flow and tax. Standard errors are heteroskedasticity robust.

Figure 9. Optimal Congestion Taxes by Time Period



Notes: Average period taxes use contemporaneous values of time. Morning peak is 6:00–8:59, afternoon peak is 15:00–17:59, and off-peak contains all other weekday hours.

E Welfare Derivation for Income-Group Taxes

The welfare accounting follows the heterogeneous-user congestion-pricing framework of [Verhoef and Small \(2004\)](#), in which demand and the value of time differ across users sharing congested roads. Here I apply that accounting to two income groups on a common network and derive the pricing conditions under the paper's demand specification.

This appendix suppresses the county, hour, and year indices for readability. Groups $g \in \{L, H\}$ denote lower- and higher-income residential tracts, classified using the county's household-weighted 80th percentile of 2018 tract median household income. Group g has flow F_g , inverse demand $P_g(F_g) = (A_g/F_g)^{1/\eta}$ in dollars per mile, demand shifter A_g , and value of time V_g in dollars per hour. Both groups share the price-elasticity magnitude $\eta = 0.1$. Total modeled flow is $F = \sum_g F_g$, measured in passenger-car-equivalent vehicle miles per hour. Travel time $T(F)$ is in hours per mile and $T'(F)$ is its flow derivative. Each V_g equals one-half of the group's hourly earnings proxy.

Let $F_g^{ref} > 0$ be fixed reference flows and x a dummy flow variable. Welfare W , net of travel-time cost and up to an additive constant, is

$$W(F_L, F_H) = \sum_g \int_{F_g^{ref}}^{F_g} P_g(x) dx - T(F) \times \sum_g V_g \times F_g. \quad (58)$$

Positive reference flows keep these integrals finite for $\eta = 0.1$ without changing the welfare derivatives. Define $M \equiv \sum_g V_g \times F_g$, the value-of-time-weighted flow exposed to congestion. The welfare derivative for one additional vehicle mile supplied by group q is

$$\frac{\partial W}{\partial F_q} = P_q(F_q) - \underbrace{V_q \times T(F)}_{\text{private time cost}} - \underbrace{T'(F) \times M}_{\text{external congestion cost}}. \quad (59)$$

The index q denotes the group supplying the additional mile. The private term differs with the entrant's value of time. The external term does not depend on the entrant's identity: one

additional vehicle mile changes travel time for all existing users. A common monetary toll yields the private condition $P_q(F_q) = V_q \times T(F) + \tau^P$. Comparing it with Equation (59) gives

$$\tau^P = T'(F^P) \times M^P = T'(F^P) \times \sum_g V_g \times F_g^P, \quad (60)$$

where superscript P denotes the common monetary Pigouvian policy, F_g^P its group flows, $F^P = \sum_g F_g^P$, and $M^P = \sum_g V_g \times F_g^P$. The toll τ^P is in dollars per mile. This proves Equation (33). For accounting purposes, this common toll can be written as the sum

$$\tau^P = \sum_g \tau_g^{contrib}, \quad \tau_g^{contrib} = T'(F^P) \times V_g \times F_g^P. \quad (61)$$

Each contribution measures the portion of marginal external damage borne by group g . It is not the tax rate paid by an entrant from that group.

The distinction between unrestricted and constrained pricing with heterogeneous users is discussed in [Small et al. \(2024, Section 4.2.3\)](#). The particular restriction considered here is $\tau_g = V_g \times \kappa$, where τ_g is the monetary toll paid by group g in dollars per mile and κ is a common time-equivalent toll in hours per mile. The same proportionality restriction underlies the deterministic-lane-choice discount rule in [Pandey et al. \(2022, Proposition 1\)](#). Here I derive the welfare-maximizing charge for the paper's elastic group demands with a common price elasticity. The private equilibrium condition and constant-elasticity group demand are

$$P_g(F_g) = V_g \times [T(F) + \kappa], \quad F_g = A_g \times V_g^{-\eta} \times [T(F) + \kappa]^{-\eta}. \quad (62)$$

Because both groups share the same elasticity η , their flow shares do not change with a

marginal change in κ , holding the demand shifters and values of time fixed:

$$\frac{F_g}{F} = \frac{A_g \times V_g^{-\eta}}{\sum_{\ell} A_{\ell} \times V_{\ell}^{-\eta}}, \quad \frac{dF_g}{d\kappa} = \frac{F_g}{F} \times \frac{dF}{d\kappa}. \quad (63)$$

Here $\ell \in \{L, H\}$ is a summation index over the two income groups, while g denotes the group whose share is being calculated. Thus the denominator is $A_L \times V_L^{-\eta} + A_H \times V_H^{-\eta}$.

Differentiating Equation (58) along the equilibrium response to κ gives

$$\begin{aligned} \frac{dW}{d\kappa} &= \sum_g P_g(F_g) \times \frac{dF_g}{d\kappa} - T(F) \times \sum_g V_g \times \frac{dF_g}{d\kappa} - T'(F) \times M \times \frac{dF}{d\kappa} \\ &= \kappa \times \sum_g V_g \times \frac{dF_g}{d\kappa} - T'(F) \times M \times \frac{dF}{d\kappa} \\ &= M \times \frac{dF}{d\kappa} \times \left[\frac{\kappa}{F} - T'(F) \right]. \end{aligned} \quad (64)$$

The last equality uses Equation (63), which implies $\sum_g V_g \times (dF_g/d\kappa) = (M/F) \times (dF/d\kappa)$. Because $M > 0$ and $dF/d\kappa < 0$, welfare rises with the time toll below $F \times T'(F)$ and falls above it. The constrained optimum therefore satisfies

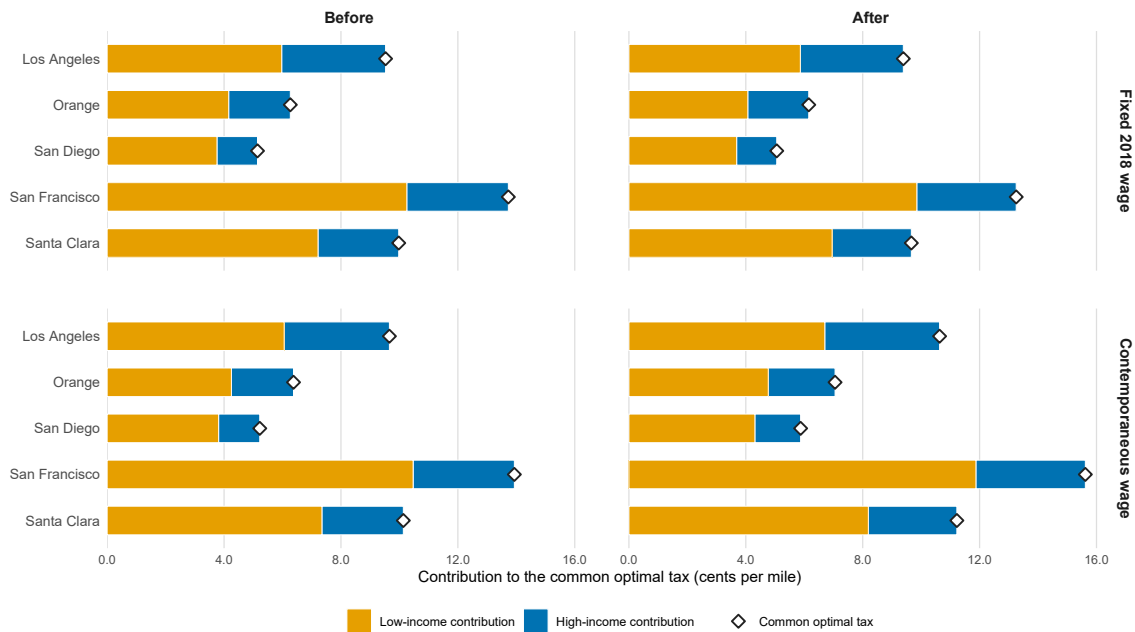
$$\kappa^* = F^{\kappa} \times T'(F^{\kappa}). \quad (65)$$

Here F^{κ} denotes total equilibrium flow under the constrained time-equivalent policy, and a star denotes its optimal time charge. This establishes Equation (36). For the Bureau of Public Roads (BPR) technology, $T(F) = (1/S^0) \times [1 + \alpha \times (F/K)^{\beta}]$, where S^0 is free-flow speed in miles per hour, K is capacity in flow units, α is the congestion scale, and β is curvature. The monetary-rate superscript V labels the value-of-time-proportional policy. Its rates satisfy

$$\kappa^* = \frac{\alpha \times \beta}{S^0} \times \left(\frac{F^{\kappa}}{K} \right)^{\beta}, \quad \tau_g^V = V_g \times \kappa^*. \quad (66)$$

The fixed-wage exercise holds V_g at its 2018 group value in every year; the contemporaneous exercise updates it using that group's year-specific wage. If group elasticities differ, Equation (63) no longer holds, and the constrained optimum must instead be obtained from the unrestricted derivative in the second line of Equation (64).

Figure 10. Income-Group Contributions to the Common Optimal Tax



Notes: Each component equals $T'(F^*) \times V_g \times F_g^*$ and the two components sum to the common monetary Pigouvian tax shown by the diamond. The components describe external-delay damages attributable to each group; they are not separate toll rates. The upper panels hold wages at 2018 levels, and the lower panels use contemporaneous wages. Before denotes 2018–2019 and after denotes 2021–2022.

F Adjusted Before–After Evidence

Let c denote county, h an observed hour, and t its calendar date, which determines the year and month. Vehicle miles traveled (VM) measure hourly traffic, Q is average speed in miles per hour, and the travel time index (TTI) is observed travel time divided by the free-flow reference time. Higher TTI means slower travel relative to that benchmark. For each outcome $Y_{cht} \in \{VM_{cht}, Q_{cht}, TTI_{cht}\}$, I estimate

$$\log Y_{cht} = \delta Post_t + \mu_{c,\omega(h,t)} + \lambda_{c,m(t)} + \varepsilon_{cht}, \quad (67)$$

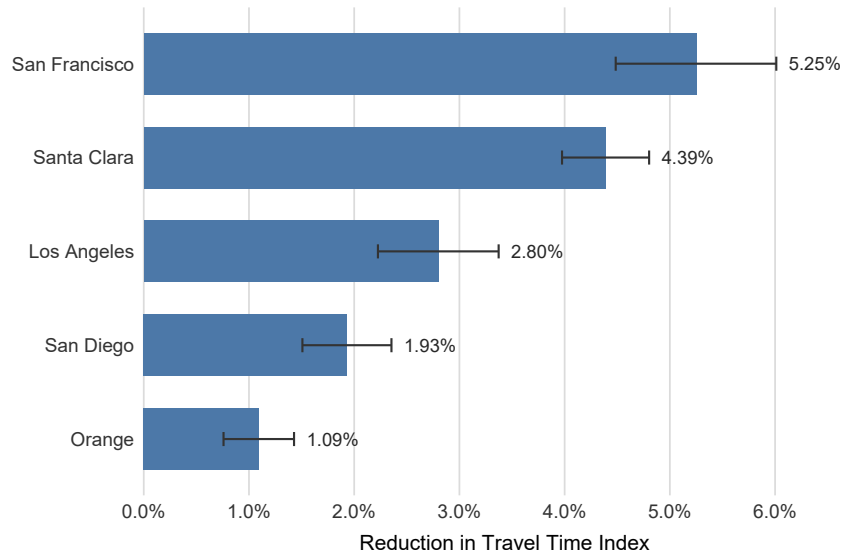
where $Post_t$ equals one in 2021–2022 and zero in 2018–2019, $\mu_{c,\omega(h,t)}$ denotes county-by-hour-of-week fixed effects, and $\lambda_{c,m(t)}$ denotes county-by-calendar-month fixed effects. The coefficient δ is the adjusted post-period change in log outcome, ε_{cht} is the regression error, $\omega(h, t)$ identifies the hour of the week, and $m(t)$ the calendar month. Here μ and λ denote the two sets of fixed effects, not demand parameters. Standard errors are clustered by year-month. The fixed effects ensure that the post-period comparison holds constant each county’s recurring weekly traffic profile and seasonal pattern. Equation (67) remains a descriptive comparison because the post indicator can capture any contemporaneous change, not only remote work.

Table 7. Adjusted Changes in Freeway Traffic and Congestion

	All hours	Afternoon peak	Weekends
<i>Panel A: Log vehicle miles traveled</i>			
Post period	-0.080*** (0.009)	-0.039*** (0.005)	-0.071*** (0.011)
Implied change (%)	-7.71	-3.87	-6.81
<i>Panel B: Log average speed</i>			
Post period	0.030*** (0.002)	0.078*** (0.006)	0.008*** (0.003)
Implied change (%)	3.08	8.07	0.76
<i>Panel C: Log Travel Time Index</i>			
Post period	-0.032*** (0.002)	-0.077*** (0.006)	-0.009*** (0.002)
Implied change (%)	-3.11	-7.44	-0.92
Observations	175,180	15,645	50,020
County $\times$ hour-of-week fixed effects	Yes	Yes	Yes
County $\times$ calendar-month fixed effects	Yes	Yes	Yes

Notes: The sample compares 2018–2019 with 2021–2022 and excludes 2020. Post period equals one in 2021–2022. Standard errors, clustered by year-month (48 clusters), are in parentheses. Afternoon peak contains weekdays from 3:00 to 5:59 p.m.; weekends contain all Saturday and Sunday hours. Implied changes equal $100[\exp(\hat{\delta}) - 1]$. The estimates are adjusted before–after comparisons and do not identify the causal effect of remote work. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 11. Adjusted Post-Pandemic Reduction in Congestion by County

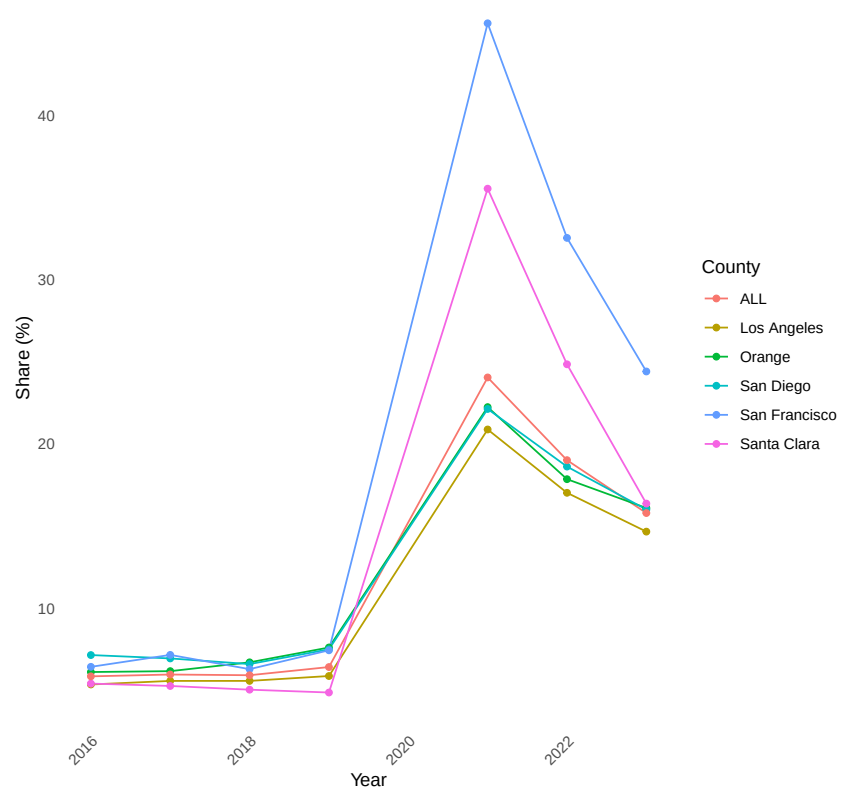


Notes: Bars report the reduction in TTI implied by county-specific versions of Equation (67). Positive values denote reductions. Whiskers are 95 percent confidence intervals based on standard errors clustered by year-month. The sample compares 2018–2019 with 2021–2022 and excludes 2020.

G Additional Descriptive Figures

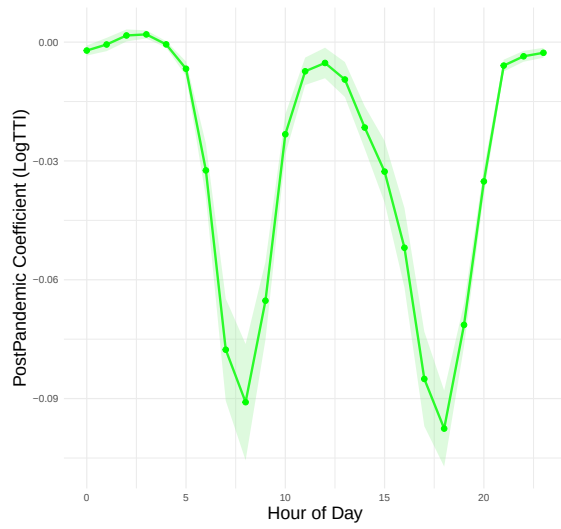
The American Community Survey (ACS) supplies the work-from-home shares. PeMS, California’s Performance Measurement System, supplies hourly freeway traffic. VMT denotes vehicle miles traveled, and TTI denotes the travel time index, observed travel time relative to the free-flow reference time. The hourly panels compare traffic and congestion profiles before and after the pandemic demand shift.

Figure 12. Work-from-Home Shares by County

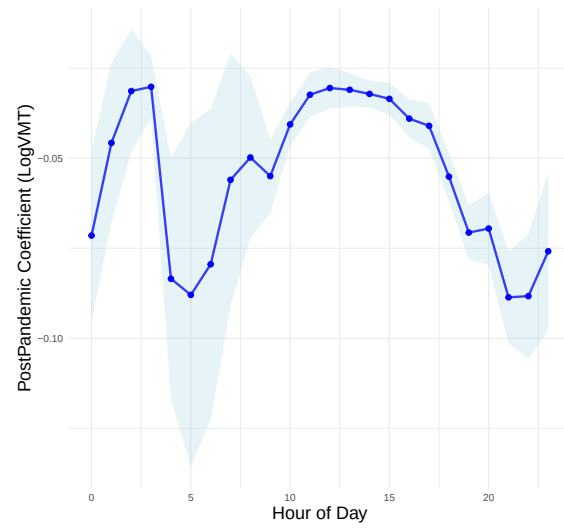


Notes: ACS work-from-home shares for the five study counties.

Figure 13. Hourly Changes in Traffic Conditions



(a) Travel Time Index



(b) Vehicle Miles Traveled

Notes: Hourly PeMS traffic profiles before and after the pandemic demand shift.